

**Department of
Veterans Affairs**

Memorandum

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Subj: Staphylococcus Aureus Bacteremia ID Consultation Policy Proposal

To: VAAAHS P&T Committee

Background

Infectious Disease specialist involvement is integral to the management of complex infections, to infection control and prevention, to promoting antimicrobial stewardship, and has been shown to improve clinical outcomes during inpatient hospitalization and outpatient care.¹ In a study from 2016 unadjusted Medicare data ID intervention of patients with ID diagnoses was associated with lower index stay mortality and, when risk adjusted, showed better outcomes and lower cost of care.²

There is clear evidence, in cases of *Staphylococcus aureus* bacteremia, that routine consultation with Infectious Disease specialists results in more frequent detection of endocarditis and metastatic infection³ and importantly, decreases all-cause mortality.⁴⁻⁶ Consultation with Infectious Diseases also leads to guideline-congruent management of patients with *Staphylococcus aureus* bacteremia.⁷

Furthermore, “bedside” ID involvement is superior to e-consult or telemedicine for *Staphylococcus aureus* bacteremia, more often leading to identification of removable sources of infection, echocardiography studies, follow-up blood cultures, and an appropriate course and duration of antimicrobial therapy.⁸

Current State

The VA Ann Arbor Healthcare System (VAAAHS) currently has inpatient Infectious Diseases consultation available in-person 7 days a week, with ID faculty and fellows reachable outside of office hours through the paging system. Primary inpatient care teams may request an ID consult at any point in the patient’s hospitalization, with routine consults seen on the same day if the consult is placed before 11 AM, or if the consult is deemed to be urgent/emergent. The vast majority of ID consults are typically seen and staffed within 24 hours of request. Antimicrobial guidelines and antibiogram data for VAAAHS are available to all physicians within the hospital. No infectious disease state currently requires a mandatory ID consult, although it is strongly suggested for gram positive bacteremia, candidemia, endocarditis, invasive fungal infection, infection in transplant patients, and HIV, among others.

Recent case prompting review:

A 66-year-old patient presented to the Ann Arbor VA Emergency Department with pain, swelling, and erythema of the right index finger. He was seen by the ED physician and Plastic Surgery, observed for 24 hours on broad spectrum antibiotics and then discharged on a course of oral antibiotics. As he was driving home from the ED he was notified that his blood cultures were positive for *Staphylococcus aureus*, but it was recommended that he continue the oral antibiotic and follow up in an outpatient plastic surgery clinic. Notably, this recommendation conflicts with several guideline-directed recommendations, including need for repeat culture, echocardiography, and a minimum of 2 weeks of IV antibiotics. When the patient did present to the clinic he required inpatient admission, was found to be in DKA, required amputation of the right index finger, and developed endocarditis of the tricuspid and pulmonic valves. While we cannot know what would have happened had the patient been instructed to return to the ED when the blood cultures were initially found to be positive, we can say that he had several negative outcomes, (endocarditis, right index finger amputation, and need for prolonged antibiotics) after he failed to receive standard of care for *Staphylococcus aureus* bacteremia.

Suggested Intervention:

As is currently the protocol, for all patients with gram positive bacteremia, the clinical microbiology laboratory will alert the primary inpatient team, ER provider, or primary outpatient provider of blood culture results (standard practice). That primary provider will still be expected to appropriately respond to the positive cultures. *In addition to the current protocol*, we will now also ask the clinical microbiology laboratory to send an FYI page to the Infectious Diseases Fellow on call through the paging website. The timing of this page should be when the organism in the blood culture is identified as *Staphylococcus aureus*. In the present workflow, this identification is made by use of the Biofire BCID2 panel, though there will be times this identification is made later in the workup, such as when MALDI-TOF is performed or with subsequent biochemical tests. It will be the responsibility of the Infectious Diseases team to ensure an Infectious Diseases consult is placed, and that the patient is seen by the Infectious Diseases team in a timely manner. Additionally, we propose to review, on a yearly basis, all of the patient with *Staphylococcus aureus* bacteremia, and whether they did or did not receive an Infectious Diseases consultation.

References

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